

Diagonalization & Matlab

Diagonalization

Four Step Method

Step 1. Find the eigenvalues of A .

Step 2. Find three linearly independent eigenvectors of A .

Step 3. Construct P from the vectors in step 2.

Step 4. Construct D from the corresponding eigenvalues.

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1: Find the eigenvalues

We solve the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

So:

$$\begin{aligned} \det \left(\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix} \right) &= (4-\lambda)(3-\lambda) - (2)(1) \\ &= (4-\lambda)(3-\lambda) - 2 = (12 - 4\lambda - 3\lambda + \lambda^2) - 2 = \lambda^2 - 7\lambda + 10 \end{aligned}$$

Solve:

$$\lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$$

So the eigenvalues are:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

Step 2: Find the eigenvectors

For $\lambda = 5$:

Solve $(A - 5I)\mathbf{v} = 0$:

$$A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Row reduce or observe:

$$-1x + 1y = 0 \Rightarrow x = y$$

So eigenvector is any multiple of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda = 2$:

$$A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \Rightarrow 2x + y = 0 \Rightarrow y = -2x$$

So eigenvector is any multiple of $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Step 3: Form matrices P and D

Use the eigenvectors as **columns** of P :

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

So we have:

$$A = PDP^{-1}$$

You can check this by computing P^{-1} , then confirming $PDP^{-1} = A$

Summary

- $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$
- Diagonalizable: YES
- Eigenvalues: 5 and 2
- Eigenvectors: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}$
- Diagonal form:

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

Let's **verify the diagonalization** by computing P^{-1} , then checking whether:

$$A = PDP^{-1}$$

We already have:

- $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$
- $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$
- $D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$

Step 1: Compute P^{-1}

Use the formula for the inverse of a 2×2 matrix:

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

For $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$:

- $\det(P) = (1)(-2) - (1)(1) = -2 - 1 = -3$

So:

$$P^{-1} = \frac{1}{-3} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

Step 2: Compute PD

First compute PD :

$$PD = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} (1)(5) + (1)(0) & (1)(0) + (1)(2) \\ (1)(5) + (-2)(0) & (1)(0) + (-2)(2) \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}$$

Now compute PDP^{-1} :

$$PDP^{-1} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

Let's multiply:

First row:

- Entry (1,1): $5(2/3) + 2(1/3) = 10/3 + 2/3 = 12/3 = 4$
- Entry (1,2): $5(1/3) + 2(-1/3) = 5/3 - 2/3 = 3/3 = 1$

Verified!

Second row:

- Entry (2,1): $5(2/3) + (-4)(1/3) = 10/3 - 4/3 = 6/3 = 2$
- Entry (2,2): $5(1/3) + (-4)(-1/3) = 5/3 + 4/3 = 9/3 = 3$

Conclusion:

We have successfully diagonalized A , and confirmed:

So:

$$A = PDP^{-1}$$

$$PDP^{-1} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A$$

Diagonalization

EXAMPLE Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

SOLUTION

Step 1. Find the eigenvalues of A .

$$\begin{aligned} 0 &= \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4 \\ &= -(\lambda - 1)(\lambda + 2)^2 \end{aligned}$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$.

Diagonalization

SOLUTION (Continued...)

Step 2. Find three linearly independent eigenvectors of A .

Three vectors are needed because A is a 3×3 matrix. This is the critical step. If it fails, then A cannot be diagonalized.

$$\text{Basis for } \lambda = 1: \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{Basis for } \lambda = -2: \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

You can check that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set.

Diagonalization

SOLUTION (Continued...)

Step 3. Construct P from the vectors in step 2.

$$P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Note:

The order of the vectors is unimportant.

Diagonalization

SOLUTION (Continued...)

Step 4. Construct D from the corresponding eigenvalues.

Use the eigenvalue $\lambda = -2$ twice, once for each of the eigenvectors corresponding to $\lambda = -2$:

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Note: It is essential that the order of the eigenvalues match the order chosen for the columns of P .

Diagonalization

SOLUTION (Continued...)

Verification:

It is a good idea to check that P and D really work. To avoid computing P^{-1} , simply verify that $AP = PD$. This is equivalent to $A = PDP^{-1}$ when P is invertible. (However, be sure that P is invertible!) Compute

$$AP = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$$PD = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$



Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix}$$

Find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

$$\det \left(\lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix} \right) = 0$$

This computation is left as an exercise, and you should verify that the eigenvalues are $\lambda_1 = 2, \lambda_2 = 2$, and $\lambda_3 = 6$.

Next, we need to find the eigenvectors. We first find the eigenvectors for $\lambda_1, \lambda_2 = 2$. Solving $(2I - A)X = 0$ to find the eigenvectors, we find that the eigenvectors are

$$t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

where t, s are scalars. Hence there are two basic eigenvectors which are given by

$$X_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

You can verify that the basic eigenvector for $\lambda_3 = 6$ is $X_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$

Then, we construct the matrix P as follows.

$$P = [X_1 \quad X_2 \quad X_3] = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

That is, the columns of P are the basic eigenvectors of A . Then, you can verify that

$$P^{-1} = \begin{bmatrix} -\frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$$

Thus,

$$\begin{aligned} P^{-1}AP &= \begin{bmatrix} -\frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & -1 \\ -2 & -4 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix} \end{aligned}$$

Diagonalization

EXAMPLE

Determine if the following matrix is diagonalizable.

$$A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

SOLUTION This is easy! Since the matrix is triangular, its eigenvalues are obviously 5, 0, and -2 . Since A is a 3×3 matrix with three distinct eigenvalues, A is diagonalizable.

Eigenvalues of a triangular matrix. It is easy to compute the determinant of an upper- or lower-triangular matrix; this makes it easy to find its eigenvalues as well.

How diagonalization helps in Markov chains.

A **Markov chain** is a system that transitions between states over time, where the **next state depends only on the current state**, not on the history.

It is described using a **transition matrix** P :

- Rows = current state
- Columns = next state
- Entries = probabilities
- Each row sums to 1

Suppose we have **two states**:

- State 1 = Sunny
- State 2 = Rainy

Transition matrix:

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

- If it's Sunny today (state 1), there's a 90% chance it'll be Sunny tomorrow, 10% chance Rainy.
- If Rainy today (state 2), 50% chance for each tomorrow.

Predict the **weather several days later**, e.g. P^n = probability matrix after n days.

Step-by-step: Use Diagonalization to Compute P^n

We want:

$$P^n = MD^nM^{-1}$$

Where:

- M = eigenvectors of P
 - D = diagonal matrix of eigenvalues
-

Step 1: Find Eigenvalues of P

$$\det(P - \lambda I) = \begin{vmatrix} 0.9 - \lambda & 0.1 \\ 0.5 & 0.5 - \lambda \end{vmatrix} = (0.9 - \lambda)(0.5 - \lambda) - 0.05$$

Expanding:

$$\begin{aligned}(0.9 - \lambda)(0.5 - \lambda) &= 0.45 - 1.4\lambda + \lambda^2 \\ \Rightarrow \lambda^2 - 1.4\lambda + 0.4 &= 0\end{aligned}$$

Solve:

$$\lambda = \frac{1.4 \pm \sqrt{(-1.4)^2 - 4(0.4)}}{2} = \frac{1.4 \pm \sqrt{1.96 - 1.6}}{2} = \frac{1.4 \pm \sqrt{0.36}}{2} = \frac{1.4 \pm 0.6}{2}$$

So:

- $\lambda_1 = 1$
- $\lambda_2 = 0.4$

Step 2: Find Eigenvectors

For $\lambda = 1$:

Solve $(P - I)v = 0$:

$$P - I = \begin{bmatrix} -0.1 & 0.1 \\ 0.5 & -0.5 \end{bmatrix} \Rightarrow x = y$$

Eigenvector: $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda = 0.4$:

$$P - 0.4I = \begin{bmatrix} 0.5 & 0.1 \\ 0.5 & 0.1 \end{bmatrix} \Rightarrow 5x + y = 0 \Rightarrow y = -5x$$

Eigenvector: $v_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$

Step 3: Form M , D , and M^{-1}

$$M = \begin{bmatrix} 1 & 1 \\ 1 & -5 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 0.4 \end{bmatrix}$$

Find M^{-1} :

$$\det(M) = (1)(-5) - (1)(1) = -6 \Rightarrow M^{-1} = \frac{1}{-6} \begin{bmatrix} -5 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5/6 & 1/6 \\ 1/6 & -1/6 \end{bmatrix}$$

Step 4: Compute $P^n = MD^nM^{-1}$

Since $D^n = \begin{bmatrix} 1^n & 0 \\ 0 & 0.4^n \end{bmatrix}$, we can compute P^n quickly for any n !

For large n , $0.4^n \rightarrow 0$

So:

$$P^n \approx M \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} M^{-1} = \text{steady-state matrix}$$

Step 5: Compute P^∞

$$\begin{aligned} P^\infty &= M \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} M^{-1} = (\text{first column of } M) \cdot (\text{first row of } M^{-1}) \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 5/6 & 1/6 \end{bmatrix} = \begin{bmatrix} 5/6 & 1/6 \\ 5/6 & 1/6 \end{bmatrix} \end{aligned}$$

This is the **long-term (steady-state) transition matrix**.

Step 6: Find Steady-State Distribution

Let $\pi = [\pi_1 \quad \pi_2]$, the steady-state distribution.

Solve:

$$\pi P = \pi \quad \text{and} \quad \pi_1 + \pi_2 = 1$$

From $\pi P = \pi$, we get:

$$0.9\pi_1 + 0.5\pi_2 = \pi_1 \Rightarrow -0.1\pi_1 + 0.5\pi_2 = 0 \Rightarrow \pi_1 = 5\pi_2$$

Use normalization:

$$\pi_1 + \pi_2 = 1 \Rightarrow 5\pi_2 + \pi_2 = 1 \Rightarrow 6\pi_2 = 1 \Rightarrow \pi_2 = \frac{1}{6}, \quad \pi_1 = \frac{5}{6}$$

Final Answers:

Transition Matrix:

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$$

Steady-State Vector (Long-term Probability):

$$\pi = \left[\boxed{5/6} \quad \boxed{1/6} \right]$$

This means:

- In the long run, 83.3% of the time it's **Sunny**
- 16.7% of the time it's **Rainy**

Matlab

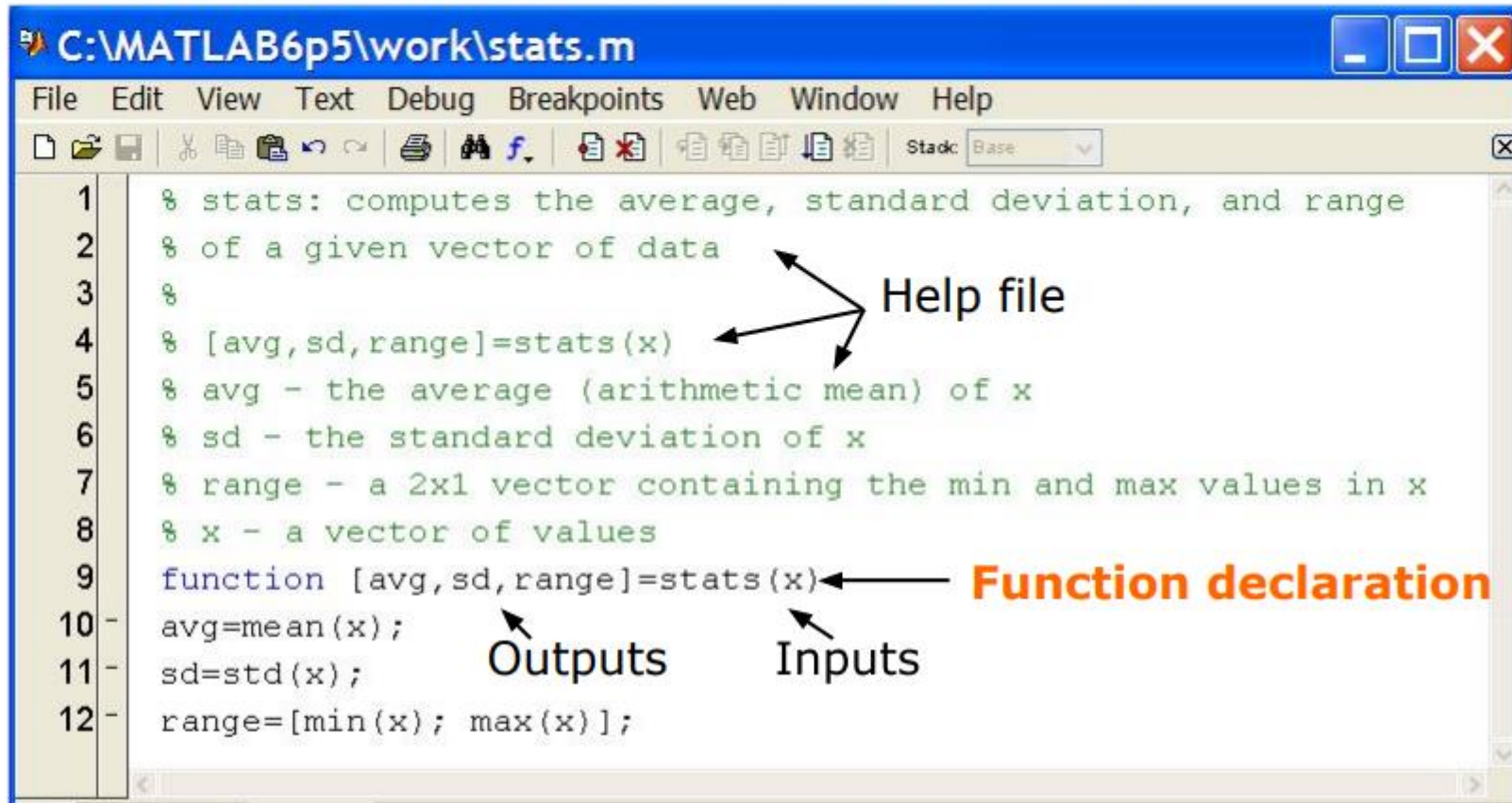
Function

The first non-commented line of a function m-file contains the function header, which is of the form : `function [out1,out2,...] = filename(arg1,arg2,...);`

The m-file ends with the command `return`, which returns the program execution to the place where the function was called.

The function code is executed whenever, either at the interactive command prompt or within another m-file, it is invoked with the command : `[outvar1,outvar2,...] = filename(var1,var2,...)`

- Functions look exactly like scripts, but for **ONE** difference
 - Functions must have a function declaration



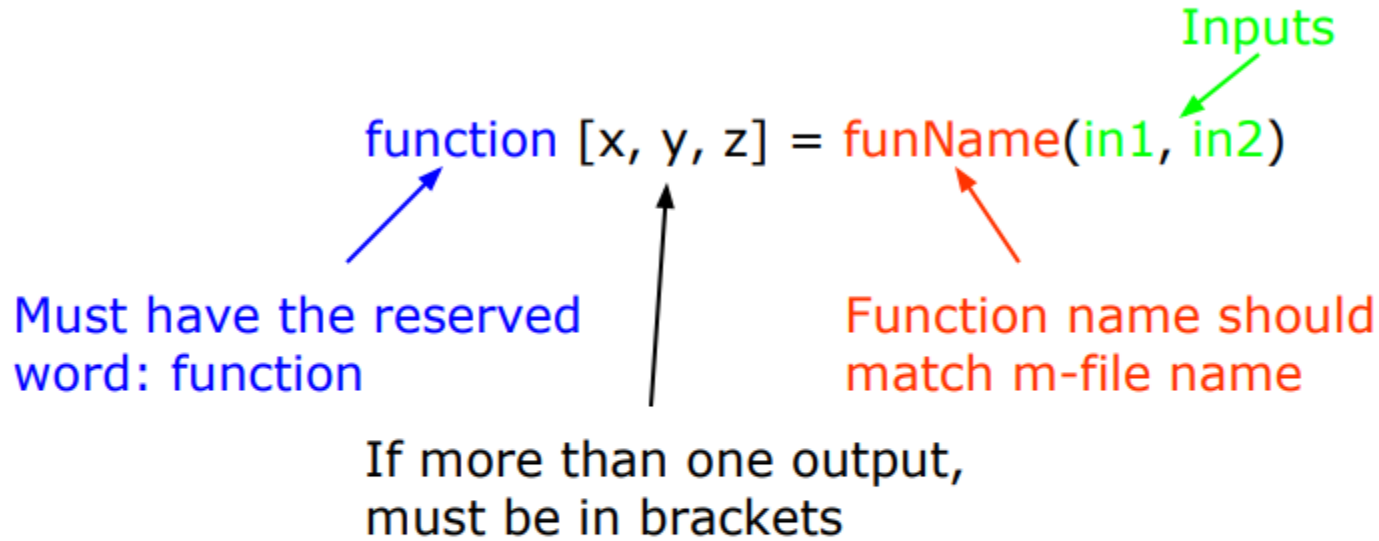
The image shows a MATLAB editor window titled "C:\MATLAB6p5\work\stats.m". The window contains a function script with the following code:

```
1 % stats: computes the average, standard deviation, and range
2 % of a given vector of data
3 %
4 % [avg,sd,range]=stats(x)
5 % avg - the average (arithmetic mean) of x
6 % sd - the standard deviation of x
7 % range - a 2x1 vector containing the min and max values in x
8 % x - a vector of values
9 function [avg,sd,range]=stats(x)
10 avg=mean(x);
11 sd=std(x);
12 range=[min(x); max(x)];
```

Annotations on the image:

- An arrow labeled "Help file" points to the function declaration line 9: `function [avg,sd,range]=stats(x)`.
- An arrow labeled "Outputs" points to the output variables `[avg,sd,range]` in the function declaration.
- An arrow labeled "Inputs" points to the input variable `x` in the function declaration.
- The text "Function declaration" is written in orange next to the function declaration line.

- Some comments about the function declaration


The diagram shows the MATLAB function declaration syntax: `function [x, y, z] = funName(in1, in2)`. Annotations include:

- A blue arrow pointing to the word `function` with the text "Must have the reserved word: function".
- A black arrow pointing to the output list `[x, y, z]` with the text "If more than one output, must be in brackets".
- A red arrow pointing to the function name `funName` with the text "Function name should match m-file name".
- A green arrow pointing to the input list `(in1, in2)` with the text "Inputs".

- **No need for return:** MATLAB 'returns' the variables whose names match those in the function declaration (though, you can use `return` to break and go back to invoking function)
- **Variable scope:** Any variable created within the function but not returned disappears after the function stops running (They're called "local variables")